

COMPUTER VISION – COMPARATIVE ANALYSIS

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Course Name: Computer Vision

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1. Detection Performance Comparison

Question: Two object detection models produce: Model A (Viola-Jones): TP=80, FP=20, FN=30; Model B (YOLO): TP=90, FP=15, FN=20. Compute precision and recall for both models and compare their performance.

Answer:

Precision = $TP / (TP + FP)$

Recall = $TP / (TP + FN)$

Model A: Precision = $80 / (80 + 20) = 80\%$. Recall = $80 / (80 + 30) = 72.73\%$.

Model B: Precision = $90 / (90 + 15) = 85.71\%$. Recall = $90 / (90 + 20) = 81.82\%$.

YOLO has higher precision and recall than Viola-Jones. Therefore, **Model B (YOLO) performs better for object detection.**

2. Motion Estimation Error Analysis

Question: Two motion estimation techniques produce displacement errors in pixels. Method A: 2, 3, 2. Method B: 3, 5, 4. Compute the average error and evaluate which method is more accurate.

Answer:

Average error of Method A = $(2 + 3 + 2) / 3 = 2.33$ pixels.

Average error of Method B = $(3 + 5 + 4) / 3 = 4.00$ pixels.

A lower displacement error indicates better accuracy. Since Method A has an average error of 2.33 pixels compared with 4.00 pixels for Method B, **Method A is more accurate.**

3. Optical Flow Magnitude Comparison

Question: Optical flow magnitudes are: Region R1: A=5, B=8; R2: A=6, B=10; R3: A=5, B=9. Compute average motion magnitude and evaluate which method captures stronger motion.

Answer:

Average magnitude of Method A = $(5 + 6 + 5) / 3 = 5.33$ pixels/frame.

Average magnitude of Method B = $(8 + 10 + 9) / 3 = 9.00$ pixels/frame.

Method B has the higher average motion magnitude. Therefore, based on the given values, **Method B captures stronger motion.**

4. Depth Estimation Comparison (Stereo vs SfM)

Question: Estimated depths are: Object A — Stereo=2.0 m, SfM=2.5 m; Object B — Stereo=3.0 m, SfM=3.8 m; Object C — Stereo=4.0 m, SfM=5.0 m. Actual depths are A=2 m, B=3 m, C=4 m. Compute absolute error for both methods and evaluate accuracy.

Answer:

Absolute error = |Estimated depth – Actual depth|.

Stereo Vision: $A=|2.0-2|=0$ m; $B=|3.0-3|=0$ m; $C=|4.0-4|=0$ m. Total/average absolute error = **0 m**.

Structure from Motion: $A=|2.5-2|=0.5$ m; $B=|3.8-3|=0.8$ m; $C=|5.0-4|=1.0$ m. Average error = $(0.5+0.8+1.0)/3 = 0.77$ m approximately.

Stereo Vision has zero error for all three objects, while SfM has non-zero errors. Therefore, **Stereo Vision is more accurate for the given data.**

5. Pattern Recognition Accuracy Comparison

Question: Traditional recognition correctly predicts 85 out of 100 samples, while Deep Learning correctly predicts 92 out of 100 samples. Compute accuracy and evaluate which system performs better.

Answer:

Accuracy = Correct Predictions / Total Samples $\times$ 100.

Traditional: $85/100 \times 100 = 85\%$.

Deep Learning: $92/100 \times 100 = 92\%$.

Deep Learning achieves 92% accuracy compared with 85% for the Traditional system. Therefore, **Deep Learning performs better** for the given recognition task.

Conclusion: The calculated performance measures show that YOLO has better precision and recall than Viola-Jones, Method A has lower motion estimation error, Method B has higher average optical-flow magnitude, Stereo Vision has lower depth error for the given data, and Deep Learning has higher recognition accuracy.